

Juan David Munoz Bolanos

Junior System Engineer

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+43 676 9115145

munozbolanos@gmail.com



Summary

Junior Engineer with research experience handling optomechatronic and medical devices from concept to prototype. Proven track record in translating physical concepts into functional prototypes using V-model systems engineering. Adept at collaborating in interdisciplinary teams to deliver high quality technical documentation for clinical and industrial applications.

Research Experience

Jan 2022 to Jun 2026	Medical University of Innsbruck <i>PhD Researcher in adaptive optics</i>	Innsbruck, Austria
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End to End System Design: Led the hardware to software implementation of multiphoton microscope, integrating laser sources, galvo scanners, and light modulators. Achieved enhancing optical image quality.

Systems Engineering: instrument development following V-model principles, managing requirements and tracking from concept to component testing.

Quality & Testing: Executed Design of Experiments (DoE) and optical simulations in Python. Applied testing and documentation protocols for optomechanical assemblies frameworks.

CAD & Prototyping: Designed custom optomechanical mounts and alignment tools using SolidWorks and FreeCAD, ensuring precise mechanical tolerances and seamless integration.

Mar 2020 to Dec 2021	Leibniz Institute of Photonic Technology <i>R&D Research Assistant</i>	Jena, Germany
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Spectrometer Testing: Testing and modifying a dispersive 1064 nm fiber probe Raman spectrometer for non-invasive human tissue characterization, handling optomechanical layout and detector calibration.

Data Analysis & ML: Developed a custom Python package (SpectraMap) to apply supervised machine learning algorithms to 2D hyperspectral Raman images, achieving high accuracy classification of tissue samples.

Technical Documentation: Authored peer reviewed publications detailing optical design parameters and calibration procedures.

Technical Competencies

Hardware	Adaptive optics	Multiphoton/confocal microscopy	Raman spectroscopy			
	Laser manipulation	Camera integration	3D CAD (SolidWorks, FreeCAD)			
Software	Python (scikit-learn, NumPy)	LabVIEW/FPGA	C++	OpenCV	Git	MATLAB

* Dark tags: expertise; light tags: knowledge.

Education

Jan 2022 to Jun 2026	Medical University of Innsbruck <i>PhD in Adaptive Optics</i>	Innsbruck, Austria
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Oct 2019 to Sep 2021	Friedrich Schiller Univ. Jena <i>Master of Science in Photonics</i>	Jena, Germany
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Aug 2012 to Aug
2018

University of Cauca
Bachelor of Physics Engineering

Popayán, Colombia

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Systems & Quality Engineering: V-Model, 8D Problem Solving, Design of Experiments.

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

- Biking/Hiking
- Bachata Dancing
- Acoustic Guitar
- Learning German

Languages

- English (Fluent)
- German (Intermediate B2)
- Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
monika.ritsch-marte@i-med.ac.at

PD Dr. Christoph Krafft
Leibniz IPHT
christoph.krafft@leibniz-ipht.de

INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

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CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

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RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

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URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

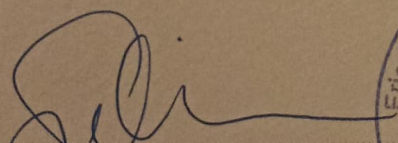
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

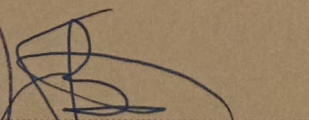
Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



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Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



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